



Cambridge International AS & A Level

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FURTHER MATHEMATICS

9231/43

Paper 4 Further Probability & Statistics

May/June 2024

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **12** pages.

- 1 A college uses two assessments, X and Y , when interviewing applicants for research posts at the college. These assessments have been used for a large number of applicants this year.

The scores for a random sample of 9 applicants who took assessment X are as follows.

21.4 24.6 25.3 22.7 20.8 21.5 22.9 21.3 22.3

The scores for a random sample of 10 applicants who took assessment Y are as follows.

20.9 23.5 24.8 21.9 23.4 24.0 23.8 24.1 25.1 25.8

The interviewer believes that the population median score from assessment X is lower than the population median score from assessment Y .

Carry out a Wilcoxon rank-sum test, at the 1% significance level, to test whether the interviewer's belief is supported by the data. [7]

21.4	4	20.9	2
24.6	15	23.5	11
25.3	18	24.8	16
22.7	8	21.9	6
20.8	1	23.4	10
21.5	5	24.0	13
22.9	9	23.8	12
21.3	3	24.1	14
22.3	7	25.1	17
		25.8	19
	70		120

$$W \sim \min(R_m, m(n+m+1) - R_m)$$

$$\sim (70, 130)$$

$$\sim 70$$

critical value $\sim 1\% \rightarrow$ one tailed, $m=9, n=10$
 $\rightarrow 61$ $70 > 61$
 $\rightarrow$ accept H_0

$$H_0: \text{med}_X = \text{med}_Y$$

$$H_1: \text{med}_X < \text{med}_Y$$





insufficient evidence to support the
interviewer's claim



- 2 A rowing club has a large number of members. A random sample of 12 of these members is taken and the pulse rate, x beats per minute (bpm), of each is measured after a 30-minute training session. A 98% confidence interval for the population mean pulse rate, μ bpm, is calculated from the sample as $64.22 < \mu < 68.66$.

- (a) Find the values of Σx and Σx^2 .

[6]

$$\bar{x} = \frac{64.22 + 68.66}{2} = 66.44$$

$$\Sigma x = 66.44 \times 12 = 797.28$$

$$66.44 \pm t_{0.99, 12-1} \sqrt{\frac{\Sigma x^2}{12}} = 64.22, 68.66$$

$$66.44 - 2.718 \sqrt{\frac{\Sigma x^2}{12}} = 64.22$$

$$\Sigma x^2 = 8.0055$$

$$\frac{1}{11} \left(\Sigma x^2 - \frac{(\Sigma x)^2}{12} \right) = 8.0055$$

$$\Sigma x^2 = 53059$$

- (b) State an assumption that is necessary for the confidence interval to be valid.

[1]

Population is normally distributed



- 3 There are three bus companies in a city. The council is investigating whether the buses reliably arrive at their destination on time. The results from random samples of buses from each company are summarised in the following table.

$$\text{exp values} = \frac{R_T \times C_T}{T}$$

		Bus company			Total
		A	B	C	
Arrival	Early	22	22	10	54
	On time	30	52	42	124
	Late	28	26	18	72
Total		80	100	70	250

Test, at the 5% significance level, whether the reliability of buses is independent of bus company. [7]

table of expected values

	A	B	C	H_0 : independence b/w reliability and company
E	17.28	21.6	15.12	
O	39.68	49.6	34.72	
L	23.04	28.8	20.16	

$$\text{Test stat} = \sum \frac{(O - E)^2}{E}$$

H_1 : non-independence

$$= \frac{(17.28 - 22)^2}{17.28} + \dots + \frac{(20.16 - 18)^2}{20.16}$$

$$= 1.289 + 0.00727 + 1.734 + 2.361 + 0.116 + 1.526 + 1.068 + 0.272 + 0.231 = 8.604$$

$$\chi^2_{df} (0.95) = 9.488 \quad \text{d.f.} = (r-1)(c-1) = 4$$

$$8.61 < 9.488 \rightarrow \text{accept } H_0$$





- 4 The random variable X has probability generating function $G_X(t)$ given by

$$G_X(t) = ct(1+t)^5,$$

where c is a constant.

- (a) Find the value of c .

[1]

$$ct(1+t)^5 \rightarrow G_X(t) \quad G_X(1) = 1$$

$$c(2)^5 = 1$$

$$c = \frac{1}{32}$$

- (b) Find the value of $E(X)$.

[2]

use product rule to find $G'_X(t)$

$$\rightarrow G'_X(t) = uv' + vu'$$

$$= \frac{1}{32}(1+t)^5 + \frac{5}{32}t(1+t)^4$$

$$G'_X(1) = E(X) = \frac{1}{32}(2)^5 + \frac{5}{32}(2)^4$$

$$= 3.5$$



The random variable Y is the sum of two independent values of X .

- (c) Write down the probability generating function of Y and hence find $\text{Var}(Y)$.

[4]

$$G_Y(t) = (G_X(t))^2$$

$$= \frac{1}{32} t^2 (t+1)^{10}$$

$$G'_Y(t) = \frac{1}{512} t (t+1)^{10} + \frac{10}{1024} t^2 (t+1)^9$$

$$G''_Y(t) = \frac{1}{512} (t+1)^{10} + \frac{10}{512} t (t+1)^9$$

$$+ \frac{10}{512} t (1+t)^9 + \frac{90}{1024} t^2 (t+1)^8$$

$$\text{Var}(Y) = G''_Y(1) + G'_Y(1) - (G'_Y(1))^2$$

$$= 44.5 + 7 - (7)^2$$

$$= \boxed{2.5}$$

- (d) Find $P(Y=5)$.

[2]

find t^5 term $\rightarrow \frac{1}{32} t^2 (1+t)^{10}$

$$\frac{10}{3} \times t^3 \times 1^7 = 120 t^3$$

expand to find t^3

$$\frac{1}{32} \times t^2 \times 120 t^3 = P(Y=5)$$

when $t=1$

$$\frac{15}{128} t^5 \rightarrow \text{so } P(Y=5) = \frac{15}{128}$$



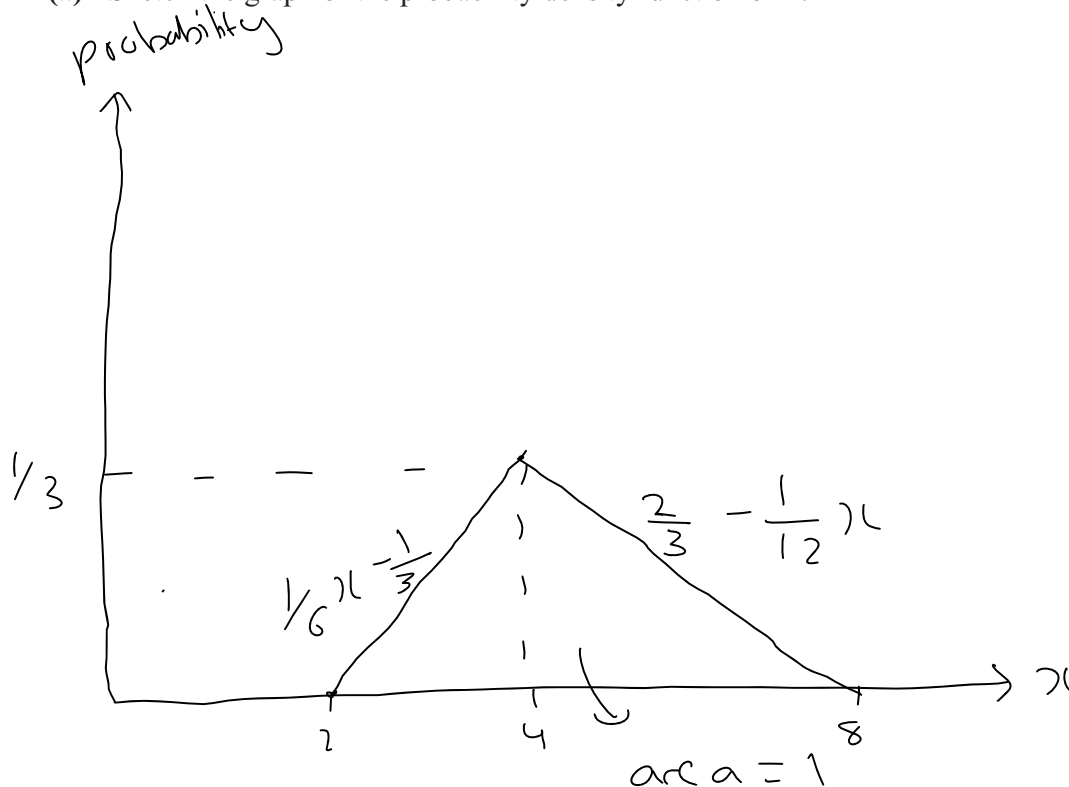


5 The continuous random variable X has cumulative distribution function F given by

$$F(x) = \begin{cases} 0 & x < 2, \\ \frac{(x-2)^2}{12} & 2 \leq x < 4, \\ 1 - \frac{(8-x)^2}{24} & 4 \leq x \leq 8, \\ 1 & x > 8. \end{cases}$$

$$\begin{aligned} & \int(x) \\ & \rightarrow \frac{2(x-2)^2}{12} = \frac{1}{6}x - \frac{1}{3} \\ & \rightarrow \frac{2(8-x)^2}{24} [3] \end{aligned}$$

(a) Sketch the graph of the probability density function of X .



(b) Find $E(X)$.

[3]

$$f(x) = \begin{cases} \frac{1}{6}x - \frac{1}{3} & 2 \leq x < 4 \\ \frac{2}{3} - \frac{1}{12}x & 4 \leq x \leq 8 \end{cases}$$

$$\begin{aligned} E(X) &= \int_2^4 \left(\frac{1}{6}x - \frac{1}{3} \right) dx + \int_4^8 \left(\frac{2}{3} - \frac{1}{12}x \right) dx \\ &= \left[\frac{1}{18}x^2 - \frac{1}{3}x \right]_2^4 + \left[\frac{2}{3}x - \frac{1}{24}x^2 \right]_4^8 \\ &= \frac{10}{9} - \frac{32}{9} = \frac{42}{9} = \frac{14}{3} \end{aligned}$$

- (c) Find the exact value of the interquartile range of X .

[4]

using C.V.F

$$L.Q \rightarrow \frac{(x-2)^2}{1.2} = 0.25 \quad x = 2 + \sqrt{3}$$

fits in range
 $2 \leq x \leq 4$

$$U.Q \rightarrow \frac{-(8-x)^2}{2.4} = 0.75$$

$$(8-x)^2 = 6$$

$$x = 8 - \sqrt{6}$$

→ in range
 $4 \leq x \leq 8$

$$\text{so IQR} = (8 - \sqrt{6}) - (2 + \sqrt{3})$$

$$\boxed{6 - \sqrt{6} - \sqrt{3}}$$



- 6 Seva is investigating the lengths of the tails of adult wallabies in two regions of Australia, X and Y . He chooses a random sample of 50 adult wallabies from region X and records the lengths, x cm, of their tails. He also chooses a random sample of 40 adult wallabies from region Y and records the lengths, y cm, of their tails. His results are summarised as follows.

$$\Sigma x = 1080 \quad \Sigma x^2 = 23480 \quad \Sigma y = 940 \quad \Sigma y^2 = 22220$$

It cannot be assumed that the population variances of the two distributions are the same.

- (a) Find a 90% confidence interval for the difference between the population mean lengths of the tails of adult wallabies in regions X and Y . [6]

$$C.I. = (\bar{x} - \bar{y}) \pm z_{\frac{\alpha}{2}} \sqrt{\frac{s_x^2}{n_x} + \frac{s_y^2}{n_y}}$$

$$z_{\frac{\alpha}{2}} = z_{0.05} = 1.645$$

$$\bar{x} = \frac{1080}{50} = 21.6 \quad \bar{y} = \frac{940}{40} = 23.5$$

$$s_x^2 = \frac{1}{49} \left(23480 - \frac{1080^2}{50} \right) = 3.102$$

$$s_y^2 = \frac{1}{39} \left(22220 - \frac{940^2}{40} \right) = 3.333$$

$$= 282 / 88$$

$$C.I. = (23.5 - 21.6) \pm 1.645 \sqrt{\frac{3.102}{50} + \frac{3.333}{40}}$$

$$= 1.9 \pm 1.645 (0.3813)$$

$$= [1.27, 2.53]$$



The population mean lengths of the tails of adult wallabies in regions X and Y are μ_X cm and μ_Y cm respectively.

- (b) Test, at the 10% significance level, the null hypothesis $\mu_Y - \mu_X = 1.1$ against the alternative hypothesis $\mu_Y - \mu_X > 1.1$. State your conclusion in the context of the question. [4]

$$Z = \frac{(\bar{y} - \bar{x}) - (\mu_Y - \mu_X)}{\sqrt{\frac{s^2_X}{n_X} + \frac{s^2_Y}{n_Y}}}$$

$$\sqrt{\frac{s^2_X}{n_X} + \frac{s^2_Y}{n_Y}}$$

$$= \frac{1.9 - 1.1}{\sqrt{\frac{3.102}{50} + \frac{3.333}{40}}} = 2.098$$

$$\sqrt{\frac{3.102}{50} + \frac{3.333}{40}}$$

critical value $\rightarrow Z_{0.1} = 1.282$ → one tailed

$2.098 > 1.282 \rightarrow$ so reject H_0

insufficient evidence to show that the tails of wallabies in region Y are longer on average than the tails of wallabies in region X



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